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Research on Wind Turbine Fault Detection Based on CNN-LSTM

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Abstract: With the wide application of wind energy as a clean energy source, to cope with the challenge of increasing maintenance difficulty brought about by the development of large-scale wind power equipment, it is crucial to monitor the operating status of wind turbines in real time and accurately identify the specific location of faults. In this study, a CNN-LSTM-based wind motor fault detection model is constructed for four types of typical faults, namely gearbox faults, electrical faults, yaw faults, and pitch faults of wind motors, combining CNN's advantages of excelling in feature extraction and LSTM's advantages of dealing with long-time sequence data, to achieve the simultaneous detection of multiple fault types. The accuracy of the CNN-LSTM-based wind turbine fault detection model reaches 90.06%, and optimal results are achieved for the effective discovery of yaw system faults, pitch system faults, and gearbox faults, obtaining 94.09%, 96.46%, and 97.39%, respectively. The CNN-LSTM wind turbine fault detection model proposed in this study improves the fault detection effect, avoids the further deterioration of faults, provides direction for preventive maintenance, reduces downtime loss due to restorative maintenance, and is essential for the sustainable use of wind turbines and maintenance of wind turbine service life, which helps to improve the operation and maintenance level of wind farms.

Keywords: fault detection; wind turbine fault; neural network; CNN-LSTM; deep learning



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1. Introduction

As a clean and renewable energy source, the development and utilization of wind energy plays an important role in addressing the global energy crisis and climate change. Wind power has become an important mode of power generation that countries worldwide are competing to develop due to the advantages of abundant reserves and low cost [1]. In recent years, with the progress of technology and policy support, the global wind power market has developed rapidly. Especially in China, the wind power market shows a strong growth trend and the total installed capacity of wind power ranks first in the world. Currently, wind power generation technology is relatively mature and easy to develop commercially on a large scale at a relatively low cost [1]. China is rich in wind energy resources and has a bright future for wind power generation [2]. According to data released by the National Energy Administration, by the end of 2023, China's installed capacity of wind power will be about 440 million kilowatts (KW), a year-on-year growth of 20.7 percent; and in 2023, the annual new wind power installed capacity will reach 75.9 million kW on the grid, a year-on-year growth of 102 percent.

As the key equipment for converting wind energy into electricity, the performance reliability and stability of wind turbines directly affect the efficiency and economy of the whole wind power system operation. Most of the wind turbines are located in areas with open terrain and high wind speeds, and the natural conditions in these areas are

conductive to the capture of wind energy, but they also create many challenges to the operation of wind turbines. Given the frequent changes in wind speed and direction in the region, the randomness is strong, the type and degree of wind motor failure vary, and the common failures are usually gearbox failure, electrical failure, yaw failure, and pitch failure four categories.

The sudden failure of wind turbine, on the one hand, will lead to production interruption and maintenance cost increase, thus bringing economic losses to the enterprise; on the other hand, there are also safety hazards, such as the wind turbine vibration exceeding the standard, which may be harmful to personal health. From the perspective of wind turbine maintenance management, given the inconvenience of wind turbine maintenance, the simplest way to reduce labor costs and accident rates is to detect the operation status in real time, take preventive maintenance measures in time, and reduce the probability of failure of wind turbines [3].

The condition detection of wind turbine operating system parameters relies on a robust fault detection system [4]. Several studies have focused on implementing a single fault detection, with most algorithmic models focusing only on the fault detection of specific wind turbine components, which mainly include the bearings and gearboxes [5–7], the rotor and blades [8,9], the pitch system [10], and the generator [11]. For example, Kusiak and Verma [10] proposed a data-based diagnostic method to monitor wind turbine pitch system faults, and Yichuan Fu et al. [12] proposed the use of the Fast Fourier Transform (FFT) and uncorrelated multilinear principal component analysis techniques for wind turbine sensor and actuator fault detection. Colombo et al. [13] proposed a sliding mode control method to adjust the pitch angle of a wind turbine operating at critical wind speed to monitor the rotor of the wind turbine; Adrian Stetco et al. [14] used various machine learning methods to achieve blade fault detection; and Yun Kong et al. [15] proposed an intelligent identification based on an Enhanced Sparse Representation-based Intelligent Recognition (ESRIR) method to achieve the detection of the bearing faults of the wind turbine. The above studies, despite their remarkable detection accuracy, are limited to the detection of a single fault, which makes it difficult to meet the demand for multi-fault detection in modern wind farms.

Based on this, different from the research idea that only focuses on a certain fault presented in previous studies, this study proposes a wind turbine fault detection method based on the Convolutional Neural Network Long Short-Term Memory (CNN-LSTM) model, which integrates four types of common faults, namely gearbox fault, electrical fault, yaw fault, and pitch fault of wind turbines, and applies a Convolutional Neural Network (CNN) to extract the local features of the wind turbine operation data and Long Short-Term Memory (LSTM) to capture the long-term dependence of the time-series data, so that the wind turbine operation condition can be monitored in real time; a multi-fault diagnosis with unbalanced samples is achieved. The operational status of the wind turbine achieves the multi-fault diagnosis of wind turbine faults with non-equilibrium samples, the early detection of fault conditions, and accurate fault localization to avoid the in-depth development of faults and take corresponding preventive maintenance measures in time, which is crucial for the sustainability of the wind turbine, the reduction in fault downtime, and the extension of its service life [16,17], as well as providing a basis of judgment for the management and maintenance of the wind turbine. This paper consists of five sections: Section 2 reviews the current research status of this study, Section 3 describes the technical route and methodology, Section 4 analyses the prediction effect of the model based on real datasets, and Section 5 describes the conclusion of the study.

2. Related Research

2.1. Research on Sensor-Based Fault Detection Methods

Early fault detection mainly relies on manual inspection and the visual inspection of various parts of the wind turbine with experience, seeking signs of abnormality or traces of failure, such as the observation method, the sound stick method, and so on. However, with the continuous development of sensor technology and fault detection technology, just relying on manual inspection cannot meet the multifaceted detection needs of wind turbine faults. At present, the classical detection methods are the real-time monitoring and data analysis of the working status of the equipment through a variety of sensors [18,19], monitoring parameters such as temperature, vibration, sound, and pressure to provide information support for fault detection.

Oil analysis is a method of determining equipment faults by examining the condition of the oil in the equipment, based on the principle that mechanical equipment faults are usually caused by abnormal wear due to poor lubrication [20]. This method is suitable for diagnosing bearing failures in equipment that relies on oil cooling or oil lubrication; however, it is limited in its applicability due to its susceptibility to interference from tumbling abrasive particles and its high dependence on operator experience.

The acoustic emission method, as a non-destructive detection method, detects damages such as cracks or deformation by analyzing the stress waves released by the materials in the equipment [21], which is more sensitive to early failure detection and can effectively exclude the low-frequency interference signals and capture the high-frequency signals with high signal-to-noise ratios, but the related instruments are more costly and less economical.

The temperature detection method determines the fault by comparing the temperature change before and after the equipment fault [22]. Temperature sensors are sensitive to capture anomalous signals when there are large variations in equipment load or speed. However, for smaller damages or faults such as spalling and pitting, it is difficult to achieve an effective detection due to small temperature variations.

The vibration detection method uses sensors to obtain vibration signals and extract the time–frequency domain feature information to determine the specific fault location by comparing the frequency of the fault features [23,24]. This method usually requires the installation of sensors inside the box or on the bearing housing, which can achieve the monitoring of both on-line and off-line states and provide more effective diagnosis of early faults [22].

2.2. Research on Fault Detection Methods Based on Feature Extraction

When the signal is collected by the sensor, given that the signal features are not obvious and easily disturbed, it is necessary to process and extract the signal features to further classify the type of fault [25]. Feature extraction refers to the extraction of different types of signal feature quantities generated by the machine during its operation and related to faults, and is usually performed in the time domain, frequency domain, and time–frequency domain.

The time domain analysis method directly characterizes the signals collected by the sensors, which is one of the simplest ways of feature extraction [26], and analyses the signals generated during the vibration of the machinery through the calculation of the necessary parameters, and the analysis data used above contain two types of parameters, namely quantitative and dimensionless parameters [27]. In the event of equipment failure, the detection of weak faults is more effective due to the linear correlation between the quantitative parameters and the degree of fault occurrence [28]. In the later stages of equipment failure, the dimensionless parameters are unstable, and it is necessary to combine several different parameters to identify the fault.

Frequency domain analysis is more complex than time domain analysis, but more stable, mainly from the spectrum of the signal, by analyzing the distribution of the signal in the frequency domain and the frequency of different vibration signals to achieve fault detection [29,30]. Spectrum analysis usually includes two methods, amplitude spectrum

and power spectrum; the amplitude spectrum aims to show the harmonic amplitude of different frequencies in the original vibration signal, while the power spectrum shows the distribution location of the power of the original vibration signal [31].

The time–frequency domain analysis method integrates the advantages of the time domain and frequency domain analysis methods, which can analyze the time–frequency relationship of vibration signals and extract the local detailed features of the signals to deal with non-smooth and non-linear signals [32]. For example, when the rolling bearing of the wind motor fails, the vibration signal usually has two characteristics of non-smooth and non-linear, and with the continuous change in time, the spectrum and time domain parameters of the vibration signal will also change; so, it is necessary to combine the two methods of the frequency domain and the time domain to extract all the characteristic information in the signal [33]. In practice, the commonly used high-frequency time–frequency domain analysis methods include empirical modal decomposition, short-time Fourier Transform, and other techniques [34].

2.3. Research on Neural Network-Based Fault Detection Methods

At present, deep learning is widely used in many fields such as image recognition and processing, computer vision, natural language processing [35–39], etc.; breaks the limitations of traditional fault detection methods that are limited to dealing with large-scale data mining; and is effective in feature extraction and model identification. Many scholars have solved a large number of fault identification problems based on deep learning methods [40–42], for example, Liu et al. [43] proposed a method for detecting rolling bearing faults based on Stacked Sparse Auto-encoders (SSAEs) and short-time Fourier Transform, which utilizes the principles of SSAE auto-learning and short-time Fourier signal processing techniques and other principles to select SoftMax as the fault type classifier. However, in the detection process, only with the help of deep learning algorithms on the classification of the classifier could the neural network originally mine the fault characteristics.

In recent years, many scholars have taken the automatic extraction of original data features based on deep learning as a research focus and conducted a large number of experimental studies to verify its high accuracy in the field of fault detection. For example, Wen et al. [44] proposed to achieve the fault detection of rotating machinery based on a novel convolutional neural network, which transforms the original one-dimensional signal into a two-dimensional gray-scale image, and then determines the specific causes leading to faults. Shao et al. [45] based on an improved Convolutional Deep Belief Network (CDBN) to accurately identify rolling bearing fault detection, whose deep network input is the original one-dimensional signal; López et al. [46] proposed a wind power prediction method based on an echo state network and long and short-term memory network. Lei et al. [47] proposed a fault diagnosis scheme based on LSTM networks, using multivariate time series as input data, implemented for both single and multi-sensor data. Therefore, it is evident that deep learning has great potential for application in fault detection.

3. Methodology

3.1. Flowchart of Fault Detection Method

In this study, wind turbine fault detection was divided into the following key steps: firstly, wind turbine state information was collected through Supervisory Control And Data Acquisition (SCADA) data; then, the data information was divided into two categories of normal state and typical faults after feature extraction; and finally, neural network was introduced to train the data to achieve fault detection and analysis. The wind turbine fault detection framework diagram is shown in Figure 1.

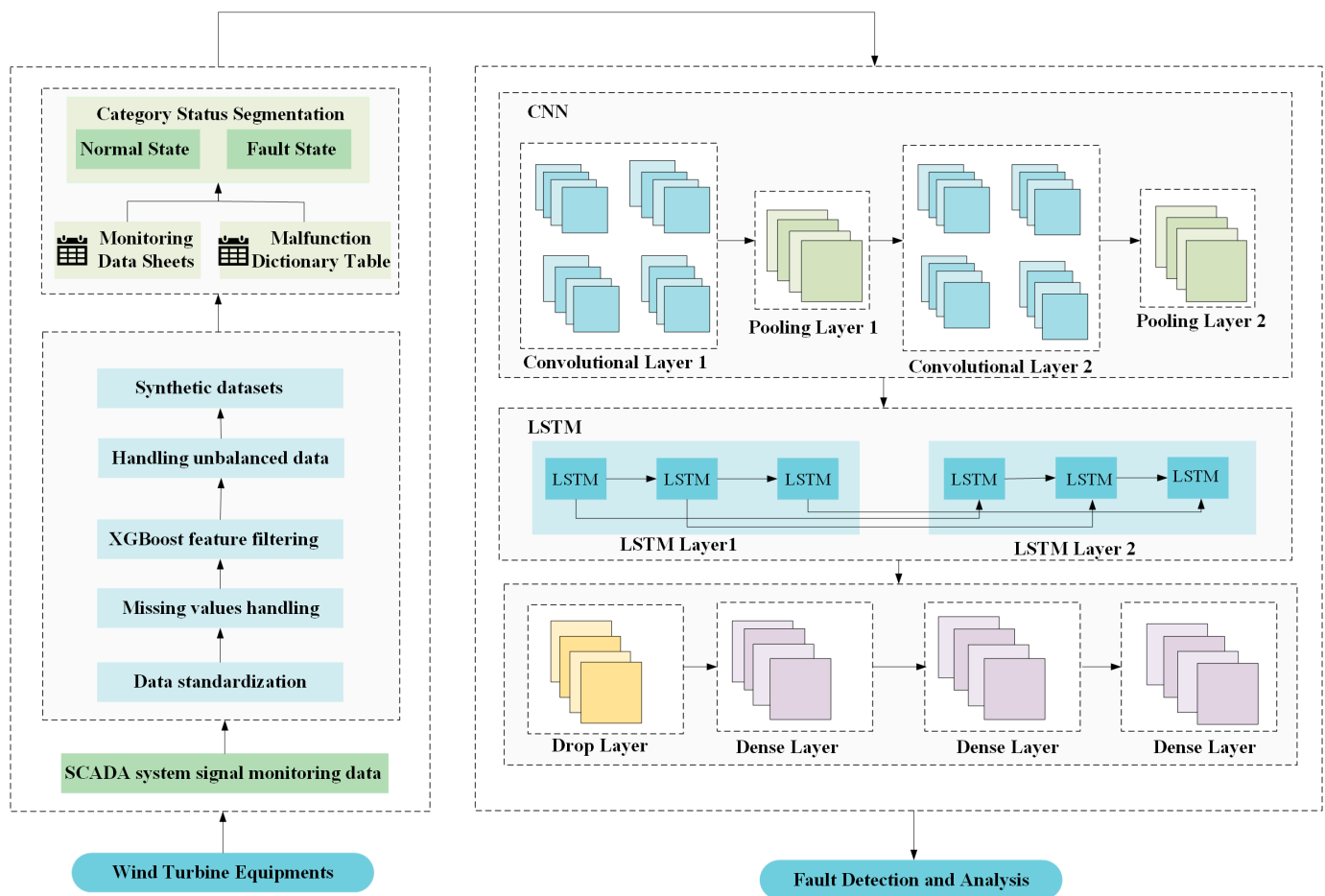


Figure 1. Framework diagram of the research idea.

3.2. Data Processing

3.2.1. Datasets

The experimental data used in this study came from the operational data of a wind farm in northeastern China obtained from a SCADA system, which is currently widely used as an important means of automatic monitoring and control of wind power plants. SCADA systems include monitoring computers, remote terminals, and logic controllers to collect and record all aspects of the operation of the wind turbine in real time. The SCADA system collects and monitors the operating parameters of the wind turbine of the wind farm, and collects more than 130 relevant operating parameters.

This study was based on real-time monitoring data from wind farms to record the changes in various indicators of wind turbines during operation. The detection status data record the operation status information every 5 min, covering the component information of the wind turbine and some environmental information, such as nacelle vibration, wind speed, yaw position, pitch speed generator speed, etc., including 133 detection categories. In addition, this study designed a fault dictionary table for wind farms, which records whether a wind turbine fails at a specific point in time and the type of failure. The time points in the fault dictionary table are not the same as the time points in the real-time monitoring data table, which are recorded based on the instantaneous time of occurrence of a fault rather than at fixed time intervals. The fault dictionary includes fault category codes, fault names, and fault levels, of which there are a total of 551 types of faults.

3.2.2. Fault Type Selection

Given the wind turbine in the operation process by the yaw error, wind speed, generator growth rate, and other elements of greater influence, the deviation will directly lead

to the wind turbine into the fault state. Therefore, it is necessary to distinguish between typical faults and other faults to make full use of the special laws of the detected object to complete the intelligent detection of equipment faults. In this study, based on the common wind turbine faults [48], and combined with the fault types presented in the experimental data, four types of typical faults, namely pitch system faults, yaw system faults, gearbox faults, and electrical system faults, were selected for the mechanism analysis, and the specific faults are summarized and analyzed in Table 1.

Table 1. Typical faults of wind turbine.

Fault Codes	System Failure Types	Specific Parts	Failure Forms	Cause of Fault
70032	Yaw system faults	Motors	Unit shutdown	Low wind speed [49]
80008	Electrical system faults	Controller	Unit shutdown	Large instantaneous speed increase [43]
160029	Pitch system faults	Bearing	Power too low	Excessive bearing fatigue [50]
70029	Gearbox faults	Converters	Overcurrent	Current sensing failure [42]

1. Yaw system faults: Common yaw system faults include different types of yaw position sensor faults, yaw position faults, yaw speed faults, and so on [49]. In the process of following the wind, due to the inability to determine the wind direction and strength of the natural wind, the yaw system needs to change constantly according to the actual situation, which makes the gears subject to alternating loads, resulting in long-term and sustained damage to the gears, affecting the wind generator cabin to accurately determine the wind direction and making it difficult for SCADA system to quickly detect and alarm.
2. Electrical system faults: The faults that often occur in the electrical system are low insulation resistance, winding short circuits, and brush slip ring failure [43]. Low insulation resistance is usually caused by the heat, sand, and dust generated by friction during the high-speed operation of bearings; winding short-circuit faults are mainly caused by friction and collision between the outside world and the internal winding of the motor.
3. Pitch system faults: Whenever the wind speed rises or falls rapidly, the pitch will be severely tested. The pitch system mainly includes faults such as the high temperature of the pitch motor and the high current of the pitch motor [50]. The excessive temperature of the pitch motor will damage the rotor and bearings of the pitch machine, and the operation overload will even cause the line aging and short circuit; the excessive current of the pitch motor will cause the internal line damage and aging, and the current information will be misreported after the failure of the current transformer, and the power of the pitch will be missing.
4. Gearbox faults: Wind turbine failure in the gearbox damage in the unit components is the highest, and gearbox disassembly and installation are more complex, with the failure of the entire wind turbine and maintenance difficulties. According to the morphology of gear damage and damage process or mechanism, the form of failure is usually divided into four categories: tooth fracture, tooth fatigue, tooth wear or scratches, and plastic deformation [42].

3.2.3. Data Preprocessing

Due to equipment operation, sampling noise, and human factors, the wind turbine operation status monitoring data have missing values, random errors, sample imbalance, large data size, and other characteristics. To ensure the effectiveness of the subsequent modeling and diagnosis, the data need to be pre-processed, specifically divided into six steps, as shown in Figure 2.

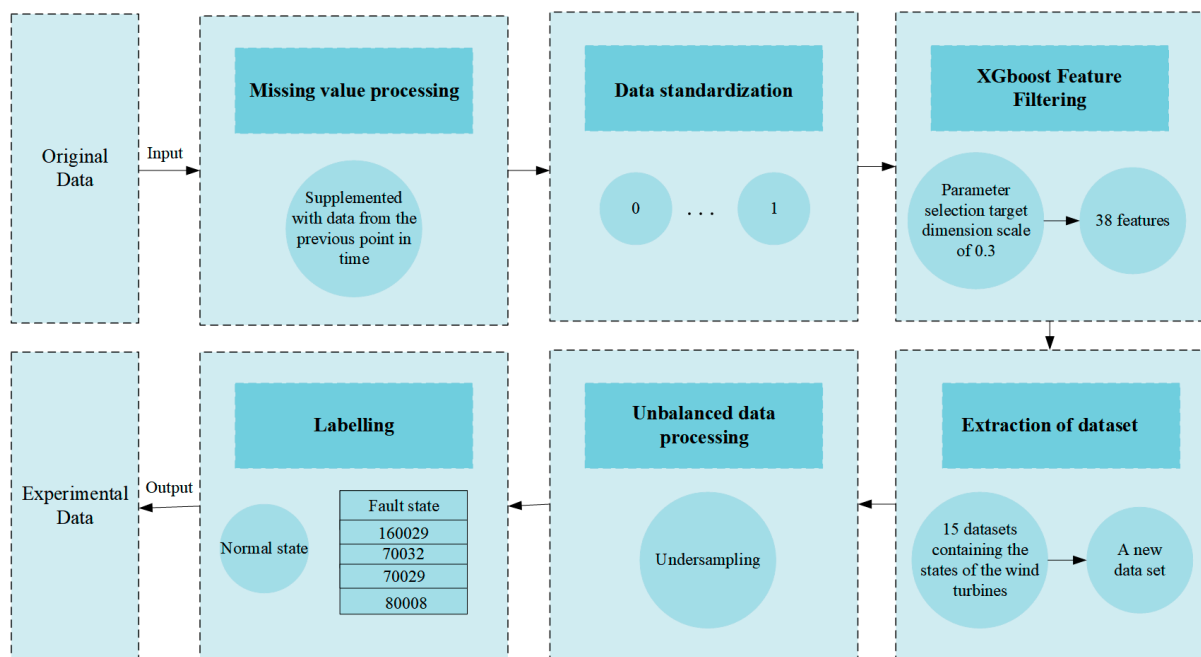


Figure 2. Flowchart of data pre-processing.

1. Missing value processing: for the occasional missing parameters caused by equipment and transmission factors, use the data of the previous moment to fill in the missing values.
2. Data normalization: using normalization methods to deal with different dimensions of the experimental data, ensure that the data distribution is between 0 and 1.
3. XGboost feature filtering: using the XGboost algorithm to select the top 30% of the original features in descending order of relevance to downscale the number of features.
4. Extraction of dataset: the condition monitoring data from 15 wind turbines of the same model are processed according to steps 1, 2, and 3, and then fused into a complete dataset.
5. Unbalanced data processing: since the normal operation samples are much more than the fault samples, the under-sampling method is used to reduce the normal samples to balance the number of samples of various types in the dataset.
6. Labeling: the textual descriptions of the operating states in the samples are transformed into the form of numerical labels, which are used to train the model.

3.3. Wind Turbine Fault Detection Based on CNN-LSTM

In this study, we used the operation data of a wind farm in northeastern China as the training set data, which records the changes in various index indices of wind turbines of models A and B within one year of operation, and selected the data of one of the wind turbines to train the model. In this study, we mainly set the number of hidden neurons, defined and initialized the input layer, and built the model network framework to regulate the CNN-LSTM-based wind turbine fault detection model [51].

3.3.1. Model Selection

The structure of the wind turbine is complex, the faults have a strong correlation with each other, and the appearance of a certain fault is often accompanied by the occurrence of other faults or the continuous change in multiple state parameters. Therefore, there is homogeneity between each feature. For example, in terms of wind speed, the wind speed of the No. 1 blade and the No. 2 blade are only measured at different locations, but the sampling principle is the same; for example, in terms of the temperature of the motor, it is based on the same sampling mechanism to obtain the parameter information at

different points in time. During the operation of the wind turbine, if there is no shutdown event, it will usually keep running continuously for a long time, and the state changes in the neighboring periods are often very subtle due to the influence of the external natural conditions and the inertia of the wind turbine itself. The operation data are usually characterized by multi-variable and multi-temporal sequences, and its operation state is strongly correlated with time.

Therefore, there is homogeneity in the characteristics of each parameter of the wind turbine's operating state, and the study of multi-failure detection of wind turbines is essentially a multi-classification process on a dataset of multivariate time series. Combining different neural network features and advantages can be derived:

1. Aiming at the existence of a local correlation between some parameter variables of wind motor, the convolutional neural network is used to achieve the aggregation of local features to carry out multi-fault detection;
2. For the global correlation between some parameter variables of the wind turbine, the full connection method is used to achieve the aggregation of global features to carry out multi-fault detection.

Therefore, in this study, the CNN-LSTM neural network model was selected for the wind turbine multi-fault detection study.

3.3.2. The Structure of CNN-LSTM Model

In the CNN-LSTM fusion network, the input data are the operating state of the wind turbine in one year, and the output data are the operating state of the wind turbine at a certain point in time, including the normal state and 4 types of typical fault states. Among them, the CNN part contains 2 convolutional layers, 2 pooling layers, 2 LSTM layers, and 3 dense layers. To improve the generalization ability of the network, the SeLU activation function is added between the convolutional layer and the pooling layer to solve the problems of gradient vanishing and overfitting, and the dropout layer is added to the output layer of the LSTM [50,52]. In Figure 3, a specific network framework is demonstrated.

Unlike traditional network models, this study used half of each LSTM and CNN model and combined them to construct a CNN-LSTM model for detecting the operation status of wind turbines at a certain point in time [53]. The selection of the model's dropout layer and activation function is described in detail below.

(1) Selection of Dropout layer

The main operations to control model overfitting include L1, L2 regular paradigm penalty term, dropout, etc. In this study, we chose to use dropout; the main reason for this choice is that dropout reduces the overfitting by randomly discarding some neurons and their bias and weights during the model training process, usually setting the percentage of the dropout as 35%, and the process does not change the loss function, and dynamically adjusting the structure of the network to reduce the overfitting and improve the generalization ability of the model. This process does not change the loss function and dynamically adjusts the network structure to reduce overfitting and improve the generalization ability of the model, which is shown in Figure 4 [54].

(2) Activation function selection

In the field of deep learning, the activation function has undergone evolution from sigmoid to more novel variants of the function such as Scaled Exponential Linear Unit (ReLU), Scaled Exponential Linear Unit (SELU), etc., among which SeLU is preferred due to its self-normalization feature in convolutional neural networks. In this study, the SeLU function was chosen as the activation function mainly because it can effectively solve

problems such as gradient vanishing and improve convergence compared to the ReLU function. The following are the specific formulas for SeLU, shown in Equation (1):

$$SeLU(x) = \lambda \begin{cases} x & x > 0 \\ \alpha e^x - \alpha & x \leq 0 \end{cases} \quad (1)$$

where λ is the scaling factor, which is used to ensure that the slope of the positive input is greater than 1, and is usually set to 1.05; and α is the negative slope parameter in the ELU function, which is used to control the saturation of the function when the negative input value is taken, and usually takes the value of 1.67.

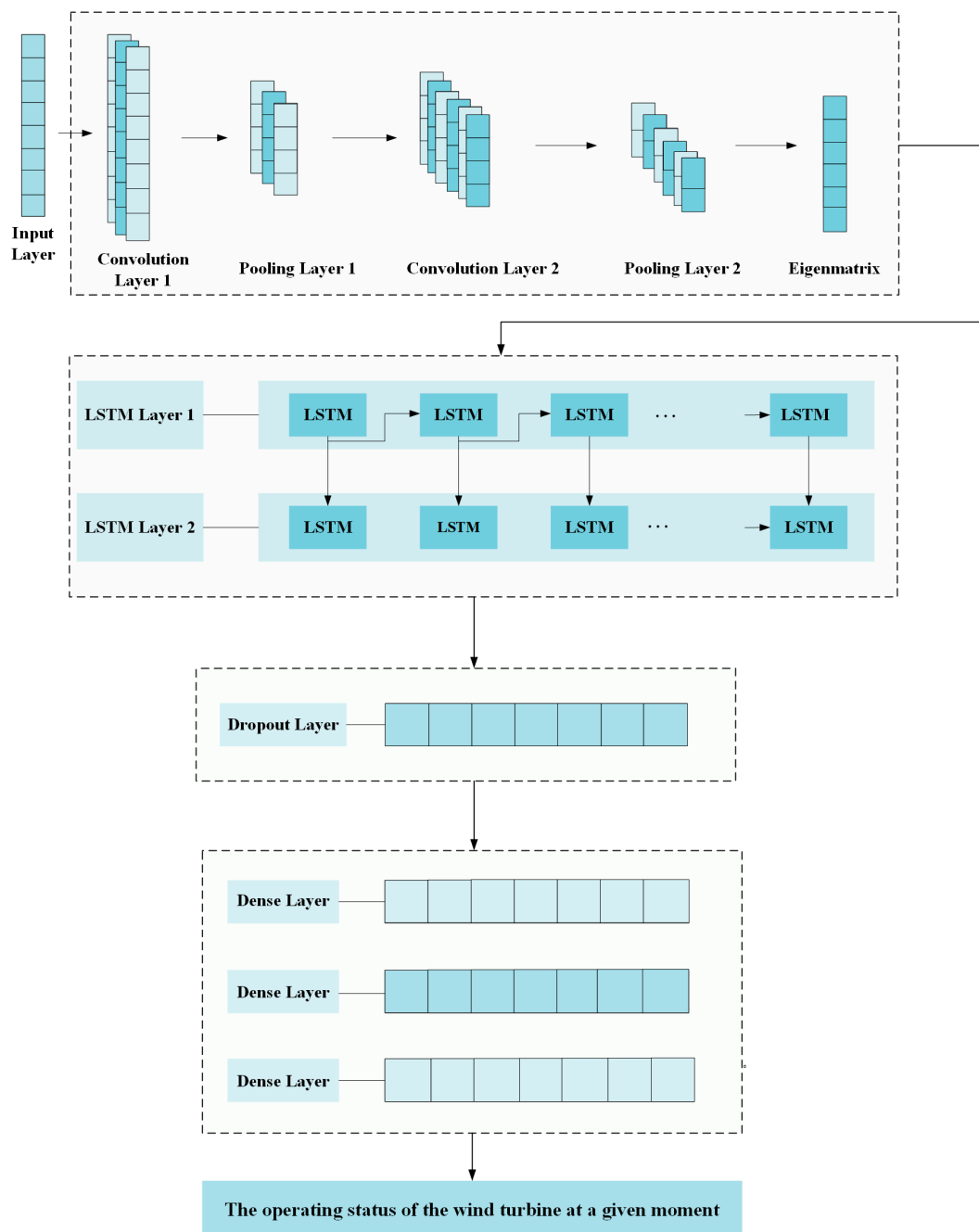


Figure 3. CNN-LSTM wind motor fault detection model architecture.

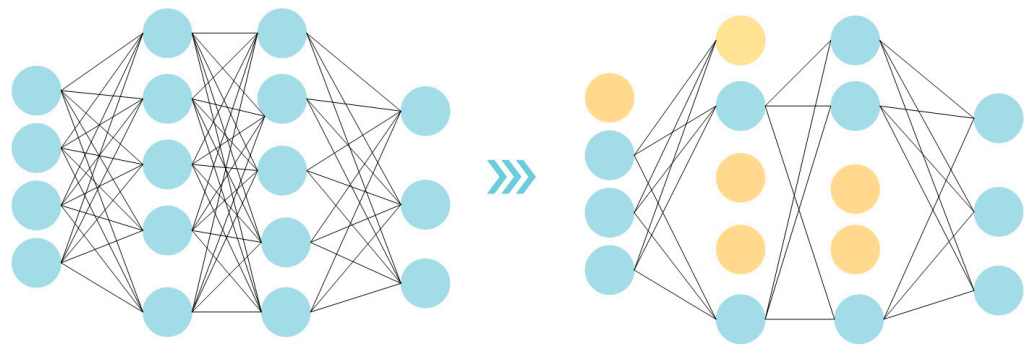


Figure 4. Schematic diagram of the dropout principle. (Neurons used in model training in blue, randomly discarded during model training in yellow).

3.3.3. Training of CNN-LSTM Fault Detection Models

To obtain the best training results, this study carried out several hyper-parameter adjustments and designed four experimental parameter configurations, and the specific adjustment results are shown in Tables 2 and 3.

Table 2. CNN-LSTM Neural Network Related Structural Framework Parameter Setting.

	Experimental Parameters	Option 1	Option 2	Option 3	Option 4
Convolution Layer_1	Convolutional kernel	32	64	32	32
	Convolution kernel size	3	2	3	3
	Activation function	SeLU	SeLU	SeLU	ReLU
Pooling Layer_1	Pooling kernel	2	3	2	2
	Convolution kernel	64	\	64	64
Convolution Layer_2	Convolution kernel size	2	\	2	2
	Activation function	SeLU	\	SeLU	ReLU
Pooling Layer_2	Pooling kernel	3	\	3	3
LSTM Layer_1	Number of neurons	100	100	40	100
	Activation function	SeLU	SeLU	SeLU	ReLU
LSTM Layer_2	Number of neurons	80	80	20	80
	Activation function	SeLU	SeLU	SeLU	ReLU
Dropout Layer	Inactivation rate	0.5	\	0.5	0.5
Dense Layer_1	Number of neurons	50	50	10	50
	Activation function	SeLU	SeLU	SeLU	ReLU
Dense Layer_2	Number of neurons	10	10	\	10
	Activation function	SeLU	SeLU	\	ReLU
Output Layer	Number of neurons	5	5	5	5
	Activation function	Softmax	Softmax	Softmax	Softmax

Table 3. CNN-LSTM Neural Network Training Parameter Scheme Selection.

	Option 1	Option 2	Option 3	Option 4
Optimizer	Adam	Adam	Adam	Adam
Batch_sizes	600	600	600	600
Epochs	500	500	500	500
Accuracy variation graph	Figure 5a	Figure 5b	Figure 5c	Figure 5d
Loss variation graph	Figure 6a	Figure 6b	Figure 6c	Figure 6d

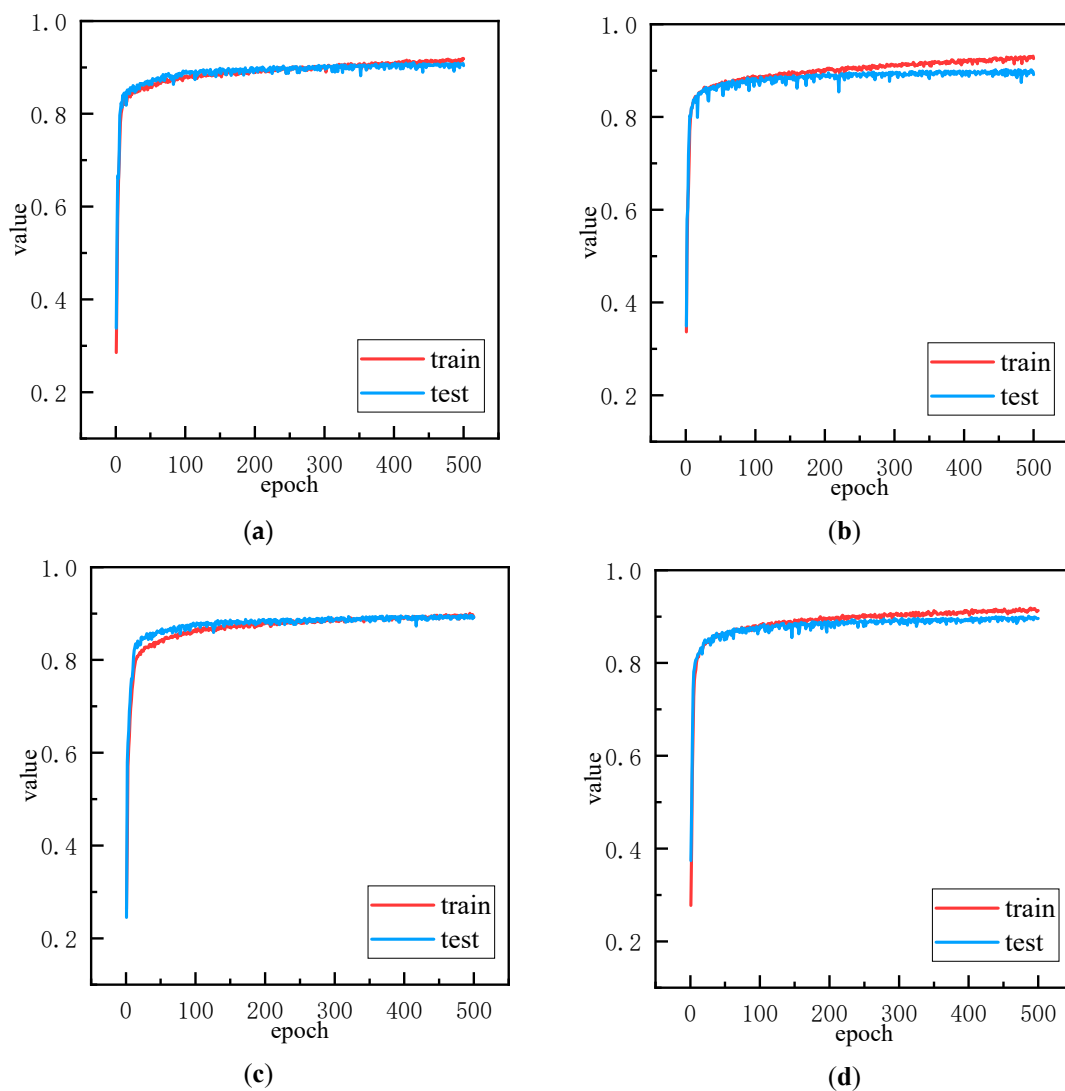


Figure 5. CNN-LSTM neural network training graph accuracy: (a) option 1; (b) option 2; (c) option 3; and (d) option 4.

As can be seen from Figures 5 and 6, with the parameter settings of the four scenarios, the training effect of scenario 1 is relatively better; so, the setting of the first one mentioned above was used in this paper when applying the CNN-LSTM neural network model for wind turbine fault detection:

1. Conv1D_1: The first convolutional layer uses 32 filters of 3×3 size, each of which learns one feature independently and performs feature mapping via the SeLU activation function.
2. MaxPooling1D: The pooling kernels of the two pooling layers are set to 2 and 3 from front to back, respectively.
3. Conv1D_2: The second convolutional layer receives the pooling output of the first layer, uses 64 filters for feature extraction, and sets the SeLU activation function for it.
4. LSTM: The second layer pooling result is taken as input, the first LSTM layer is set to 100 neurons, and the second LSTM layer is set to 80 neurons, both using the SeLU function as activation function.
5. Dropout layer: In this study, the dropout layer is set to randomly inhibit neurons with an inactivation rate of 50% to reduce the sensitivity to slight fluctuations in the data and to enhance the stability and accuracy of time series data processing.

6. Dense fully connected layer: There are three dense layers. The first layer is set with 50 neurons. The second layer is set with 10 neurons, and the third layer adopts the Softmax activation function for fault detection and outputting results.
7. batch_sizes, epochs setting values: set batch_sizes to 600 and epochs to 500.

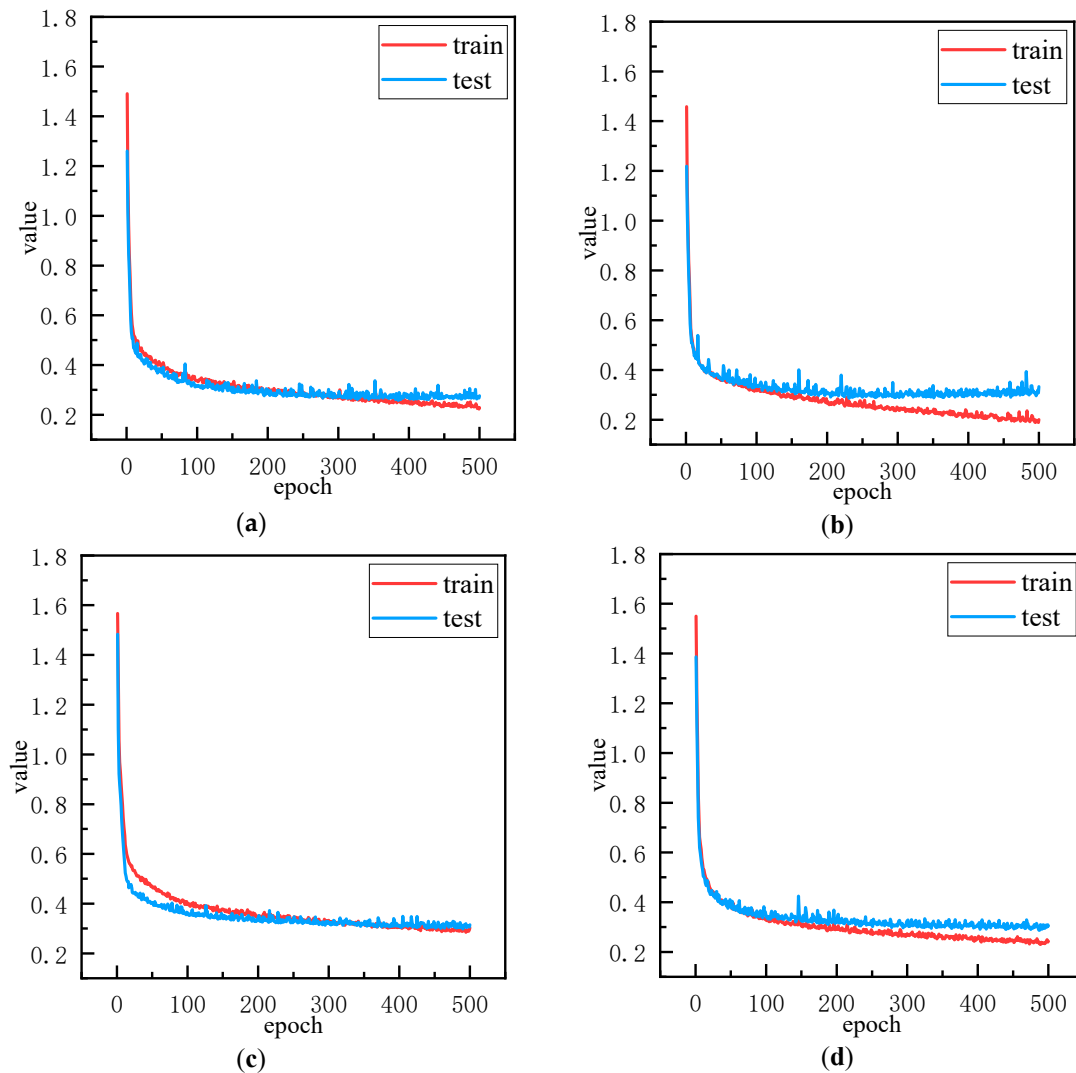


Figure 6. CNN-LSTM neural network training graph loss: (a) option 1; (b) option 2; (c) option 3; and (d) option 4.

4. Results and Discussion

4.1. Experimental Evaluation Indicators

In this study, accuracy, precision, recall, and F1 value were selected as the evaluation indexes to assess the performance of the model.

(1) Accuracy

The accuracy rate refers to the proportion of correctly detected fault samples to the total number of samples, and the formula is shown in Equation (2).

$$accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

where TP denotes a true case, i.e., the number of samples whose true labels are positive and detected as positive; FP denotes a false positive case, i.e., the number of samples whose

true labels are negative but detected as positive; TN denotes a true negative case, i.e., the number of samples whose true labels are negative and detected as negative; and FN denotes a false negative case, i.e., the number of samples whose true labels are positive but detected as negative.

(2) Precision

The precision rate for the detection results, in all the samples, was detected as positive, and the expression of the probability of actually positive samples is shown in Equation (3):

$$precision = \frac{TP}{TP + FP}, \quad (3)$$

(3) Recall

The recall rate indicates that the prediction is a positive sample, and the expression of the actual probability of positive samples is shown in Equation (4):

$$recall = \frac{TP}{TP + FN}, \quad (4)$$

(4) F1 value

To measure the performance of different types of algorithms more effectively, the F1 value is used to achieve a more accurate assessment of the algorithm performance using precision and recall as the base metrics, and the expression is shown in Equation (5):

$$f1 - score = \frac{2 \times precision \times recall}{precision + recall}, \quad (5)$$

4.2. Experimental Results

To verify the performance of the CNN-LSTM wind turbine fault detection model, this paper compared the proposed CNN-LSTM wind turbine fault detection model with the current mainstream benchmark methods CNN, LSTM, VGG16, and ResNet18 based on the same dataset, and the specific experimental results are shown in Table 4.

Through the comparison of the various types of modeling methods in Table 4, we drew the following conclusions:

1. Comprehensively comparing the precision rate, recall rate, and F1 value indexes of various types of models, the CNN model is the most ineffective;
2. In terms of effective discovery of common faults, our model achieves optimal results for yaw system faults, pitch system faults, and gearbox faults, and LSTM is only slightly better than our model in the case of power system faults. The higher recall rate for the three types of faults, namely yaw system faults, pitch system faults, and gearbox faults, indicates that our model is more sensitive to the diagnostic discovery of faults, which is crucial for the early discovery and diagnosis of faults, which is essential for avoiding the deeper development of the fault characteristics, or even causing the end of the life of the wind turbine.
3. In terms of the accuracy index, LSTM is better than our model only in the case of recognizing and detecting pitch system faults, and our model achieves the optimal value in all other cases.
4. In terms of F1 value, LSTM is better than our model only in the case of detecting the pitch system faults, but the effect of our model is also better, with only 0.48% difference from the LSTM model, and our model is better than the other models for the rest of the cases.

Table 4. CNN-LSTM neural network training parameter settings.

State	Model	Precision	Recall	f1-Score
Normal state	CNN	94.82 ± 0.23	77.84 ± 0.57	85.50 ± 0.48
	VGG16	94.91 ± 0.32	78.13 ± 0.14	85.71 ± 0.39
	LSTM	88.57 ± 0.28	84.25 ± 0.42	86.36 ± 0.41
	ResNet18	95.03 ± 0.56	78.69 ± 0.09	86.09 ± 0.47
	CNN-LSTM	95.48 ± 0.34	80.32 ± 0.61	87.25 ± 0.21
Yaw system faults	CNN	82.17 ± 0.58	92.05 ± 0.36	86.83 ± 0.45
	VGG16	83.28 ± 0.60	92.76 ± 0.13	87.76 ± 0.37
	LSTM	84.64 ± 0.54	92.21 ± 0.29	88.26 ± 0.50
	ResNet18	83.75 ± 0.43	93.44 ± 0.59	88.33 ± 0.07
	CNN-LSTM	87.13 ± 0.46	94.09 ± 0.35	90.48 ± 0.63
Electrical system faults	CNN	84.52 ± 0.20	87.61 ± 0.51	86.04 ± 0.38
	VGG16	84.94 ± 0.19	88.09 ± 0.49	86.49 ± 0.65
	LSTM	87.58 ± 0.12	88.64 ± 0.31	88.11 ± 0.52
	ResNet18	85.67 ± 0.27	88.25 ± 0.42	86.95 ± 0.55
	CNN-LSTM	89.33 ± 0.08	88.48 ± 0.44	88.90 ± 0.32
Pitch system faults	CNN	88.68 ± 0.33	91.24 ± 0.19	89.94 ± 0.64
	VGG16	88.75 ± 0.40	92.26 ± 0.55	90.47 ± 0.65
	LSTM	93.78 ± 0.11	89.15 ± 0.30	91.41 ± 0.25
	ResNet18	89.12 ± 0.48	92.61 ± 0.57	90.03 ± 0.69
	CNN-LSTM	86.37 ± 0.06	96.46 ± 0.39	90.83 ± 0.52
Gearbox faults	CNN	90.71 ± 0.28	94.54 ± 0.41	92.59 ± 0.56
	VGG16	90.28 ± 0.09	94.93 ± 0.47	92.55 ± 0.34
	LSTM	92.56 ± 0.61	94.67 ± 0.22	93.83 ± 0.46
	ResNet18	91.49 ± 0.49	95.35 ± 0.06	93.17 ± 0.39
	CNN-LSTM	93.42 ± 0.16	97.39 ± 0.29	95.36 ± 0.33

4.3. Discussion of the Results

Based on the above experimental results, it is found that different fault models perform differently for the five types of state detection. To show their respective advantages more clearly, this study conducted a comparison of four evaluation indexes: precision rate, recall rate, F1 value, and accuracy rate.

4.3.1. Performance Analysis of Fault Detection Models Based on Precision Rate

In wind turbine fault detection, the accuracy rate is used to assess whether the model has made a misdetection. For example, if the condition detected by the model at a certain point in time is not consistent with the actual condition, it means there is a misdetection. If the model has a low accuracy rate for recognizing the normal state, it may mistakenly identify the fault state as normal operation, which may lead to the real fault being ignored; if the model has a low accuracy rate for fault state detection, it may lead the staff to take wrong maintenance measures, which do not solve the faults that occur. Both of these types of misdetection situations will add risks and uncertainties to the stable operation of the wind turbine; so, the accuracy rate is a crucial indicator to assess the effectiveness of model fault detection, which directly affects the stable operation and maintenance cost of the wind turbine.

From Figure 7, we can observe that these five types of wind turbine fault detection models are more effective in identifying the normal state, which effectively reduces the possibility of identifying the faulty state as the normal state. For yaw system faults, the LSTM and CNN-LSTM models are effective, and the CNN-LSTM is better than the LSTM model, which means that capturing the long time series of wind turbine operation data features is crucial to accurately identify the detection of gearbox faults; for power system faults, our model has the best effect, with an improvement of 4.81% over CNN and 4.41% over VGG16; for pitch system faults, our model is slightly inferior to the other models, but the gap with the other models is smaller; for gearbox faults, all the five types of models

are effective, and our model is better than the gearbox faults, all five models are effective, and our model is better than the other four models. In summary, the CNN-LSTM fault detection model has the best performance in terms of accuracy.

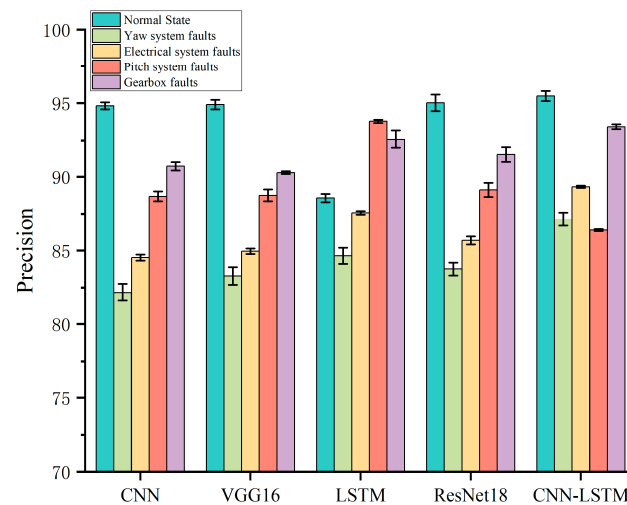


Figure 7. Comparison of the results of the precision rate of each model.

4.3.2. Performance Analysis of Fault Detection Models Based on Recall Rate

In the wind turbine fault detection task, the recall rate reflects whether the model can capture all the real fault situations, especially for the situation where the faults that are occurring are not detected, e.g., if the wind turbine has a pitch system fault at a certain point in time but the fault detection model does not detect that the pitch system fault has occurred then, it means that there is a missed detection. If the model has a low recall in a normal state, it may lead to some unnecessary preventive maintenance measures and increase the operation cost, but the threat to the stable operation of the wind turbine is small; if model has a low recall in faulty state, the wind turbine operation may miss the early fault signals, which leads to the problem not being discovered in time and even leads to the deterioration of the fault degree to the point of causing the wind turbine to shut down. Therefore, to evaluate the failure detection effect of a model, the recall rate is a relatively more important judgment index, which is directly related to whether the model can effectively improve the accuracy and timeliness of failure detection during the actual operation.

As can be seen from Figure 8, for the normal state, the LSTM model has the best effect, but the recall of the normal state has little reference value for the evaluation of the model; for the yaw system faults, the recall of each model is above 92%, and our model has the best effect, which indicates that the CNN-LSTM is very sensitive to the yaw system fault, and it can monitor the abnormal state of the yaw system promptly; for the power system faults, the recall of each model is very similar and above 87%; for the propeller system faults, our model reaches 96.61, which is 5.22% higher than the CNN, and 5.22% higher than the LSTM. For power system faults, the recall of each model is very similar and above 87%; for pitch system faults, our model reaches 96.61, which is 5.22% higher than that of CNN and 7.31% higher than that of LSTM; and for gearbox faults, our model is effective and reaches 97.39%. In summary, the CNN-LSTM fault detection model performs best in terms of recall.

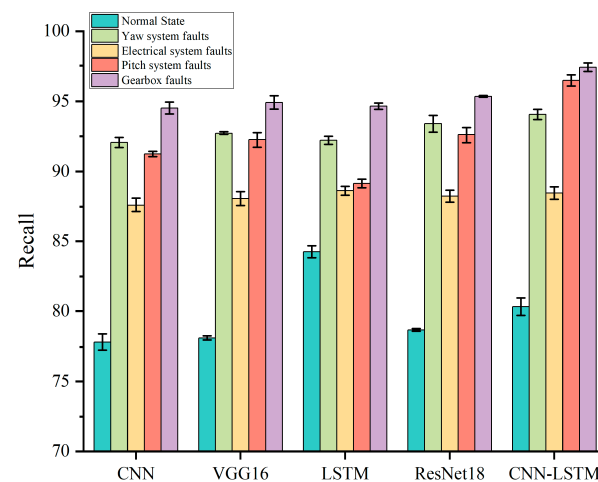


Figure 8. Comparison of the results of the recall of each model.

4.3.3. Performance Analysis of Fault Detection Models Based on F1 Value

In the wind turbine fault detection task, the F1 value serves as a key evaluation index, which comprehensively considers the overall situation of misdetection and omission detection. A lower F1 value indicates that the model has more misdetections and omissions in fault detection, and the overall effect is poor, and the opposite is superior. Both misdetection and omission situations lead to negative impacts; misdetection may lead to unnecessary maintenance or interference with normal operation, while omission may prevent potential faults from being handled promptly, which may lead to the exacerbation of or damage to the equipment's operational problems. Therefore, the F1 value, as a comprehensive metric, is a relatively important judgment indicator.

As can be seen in Figure 9, the CNN model in the identification of yaw system faults, pitch system faults, and gearbox faults detection has a poor performance for the detection of normal state and power system faults detection in general performance; for the yaw system faults and power system faults, the detection effect of each model is more than 86%, and our model detection identification of yaw system faults reached 90.48%, compared to other models, the performance of the LSTM model Improved 2.15% or more; for the pitch system faults, the LSTM model has the best performance, but the gap between our model and it is smaller; for the gearbox faults, the performance of each model is better, reaching 92% or more, and our model reaches 95.36%, which indicates that we can identify the gearbox faults more accurately and effectively avoid the situation of omission and misdetection.

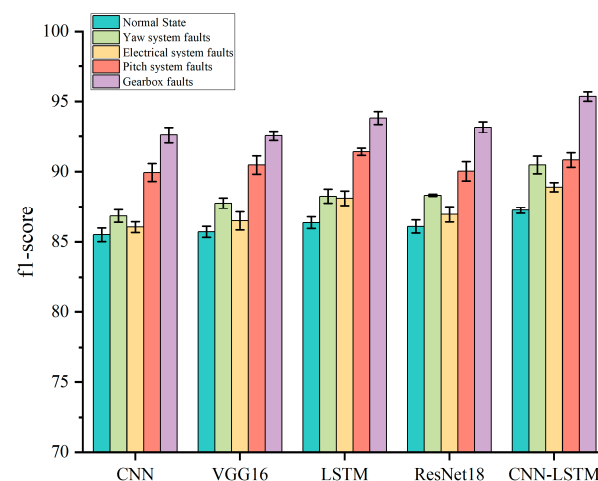


Figure 9. Comparison of results for F1 values for each model.

4.3.4. Performance Analysis of Fault Detection Models Based on Accuracy Rate

The accuracy rate applied to the wind turbine fault detection model can reflect the overall effect of the model in various aspects and is a relatively important judgment indicator. As can be seen in Figure 10, among the performances of the accuracy rate of each model, the wind turbine fault detection model based on the CNN-LSTM neural network is the most effective.

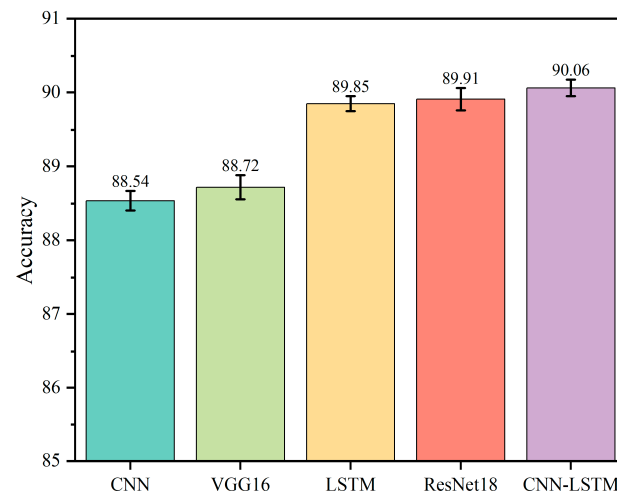


Figure 10. Comparison of accuracy results for each model.

In summary, the precision rate refers to the ratio between the number of correctly classified samples and the total number of all samples in the class, which means that the precision rate can determine the number of correct samples in the type. Recall refers to the ratio between the number of correctly categorized samples and the total number of all samples that should have been classified into that category, which means that the number of correct samples checked can be detected by the recall rate, reflecting the sensitivity of the fault detection model to various types of faults. This study found that the recall rate has a more important reference value in fault detection compared to other indicators. The F1 value is a judgment criterion to consider the precision rate and recall rate together.

From the above comparison results, it can be seen that the CNN neural network performs poorly in identifying pitch system faults and gearbox faults in terms of recall; the VGG16 fault detection model is slightly better than the CNN; the LSTM neural network performs well in dealing with long time sequence problems, improves the ability to detect long time sequences, and makes the model's performance significantly improve under the comprehensive evaluation of multiple indicators. The ResNet18, as the lightweight model, is more computationally efficient than the other indicators. A lightweight model, with high computational efficiency and effective extraction of deep features of faults, is excellent. However, the comprehensive comparison of different indicators revealed that the CNN-LSTM wind turbine fault detection model proposed in this paper is significantly better than the above algorithmic models, especially in terms of the overall high level of recall of correctly detected samples and the excellent performance of F1 values.

5. Conclusions

Given the limitations of the current expert experience-based fault prediction and single fault detection model, this study conducted an in-depth mechanistic study of typical wind turbine faults from four aspects, namely pitch system faults, yaw system faults, gearbox faults, and electrical system faults. Based on this, a wind turbine fault detection model based on CNN-LSTM was constructed, which combined the feature extraction ability of CNN and the ability of LSTM to process time series data, providing a new way of thinking in the field of wind turbine fault detection.

Our model achieved more excellent results as follows:

1. As far as the methodology is concerned, most of the previous researches focus on the identification and monitoring of a certain type of faults or the improvement in the design of a certain part of the structure of the wind turbine to extend the service life of the wind turbine. For example, Aloui R et al. [55] conducted a life cycle assessment (LCA) of the rotating part of the wind turbine to monitor the operating status to extend the service life of the operating system; Ding et al. [56] proposed a combined model for transformer state prediction to identify faults through the continuous monitoring of the transformer of the power system; and Qin et al. [57] proposed a new pitch controller strategy based on a back-propagation neural network and optimal control theory to solve the rotor speed problem to ensure the stable operation of the wind turbine. Controller strategy is used to solve the rotor speed problem to ensure the stable operation of wind motors. Our model breaks the limitation of monitoring a certain wind turbine parts and simultaneously monitors four parts, pitch system, yaw system, electrical system, and gearbox, to monitor the operation status in real time, prolong the usage time of the four systems, reduce the non-essential downtime, and prolong the service life of the wind turbine.
2. In terms of accuracy, the CNN-LSTM wind turbine fault detection model achieves an accuracy of 90.06%, which is the most effective compared to other models;
3. In terms of effective discovery of wind turbine faults, our model achieves 94.09%, 96.48%, and 97.39% recall for the identification and detection of yaw system faults, pitch system faults, and gearbox faults, respectively;
4. Considering the overall situation of misdetection and omission detection, in terms of the performance of F1 value, although the LSTM model is the most effective in identifying and detecting the pitch system faults, the gap between this model and our model is smaller and more effective, and it reaches more than 90%; in addition, our model outperforms the other four types of comparative models in various state scenarios.

Although the CNN-LSTM wind turbine fault detection model proposed in this paper outperforms the comparison model and incorporates the consideration of missing values in the model, there are still some limitations, as follows.

1. Noise interference is crucial to the effectiveness of wind turbine fault detection. The current novel fault diagnosis network based on a refined prototype and correlation weighting Manhattan distance (RPCMN) [58] and anti-noise uncertainty perception metric network [59] is effective in solving the noise problem. In the future, we will explore its application in wind turbine fault detection, which is expected to achieve better results.
2. CNN-LSTM extracts the global features of wind turbine fault monitoring data with excellent effect, but without feature dimensionality reduction, the computational cost is high. Currently, there are studies based on the neural transformer model [60], through the time domain and frequency domain to provide a more comprehensive feature representation to improve the recognition of faults at the same time to build a multi-head spatiotemporal spiking self-attention (MHSSSA) mechanism to reduce the computational cost. Reducing the computational cost is a need for the development of the fault detection field, and its future application in wind turbine multi-fault classification is expected to sustain the use of wind turbines, extend the service life of wind turbines, and better serve the wind power market.

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Abbreviations

CNN	Convolutional Neural Network
LLSTM	Long Short-Term Memory
CNN-LSTM	Convolutional Neural Network Long Short-Term Memory
kW	Kilowatt
FFT	Fast Fourier Transform
ESRIR	Enhanced Sparse Representation-based Intelligent Recognition
SSAE	Stacked Sparse Auto-encoder
CDBN	Convolutional Deep Belief Network
SCADA	Supervisory Control And Data Acquisition
ReLU	Scaled Exponential Linear Unit
SELU	Scaled Exponential Linear Unit
RPCMN	Refined Prototype and Correlation Weighting Manhattan Distance
MHSSA	Multi-head Spatiotemporal Spiking Self-attention
LCA	Life Cycle Assessment

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